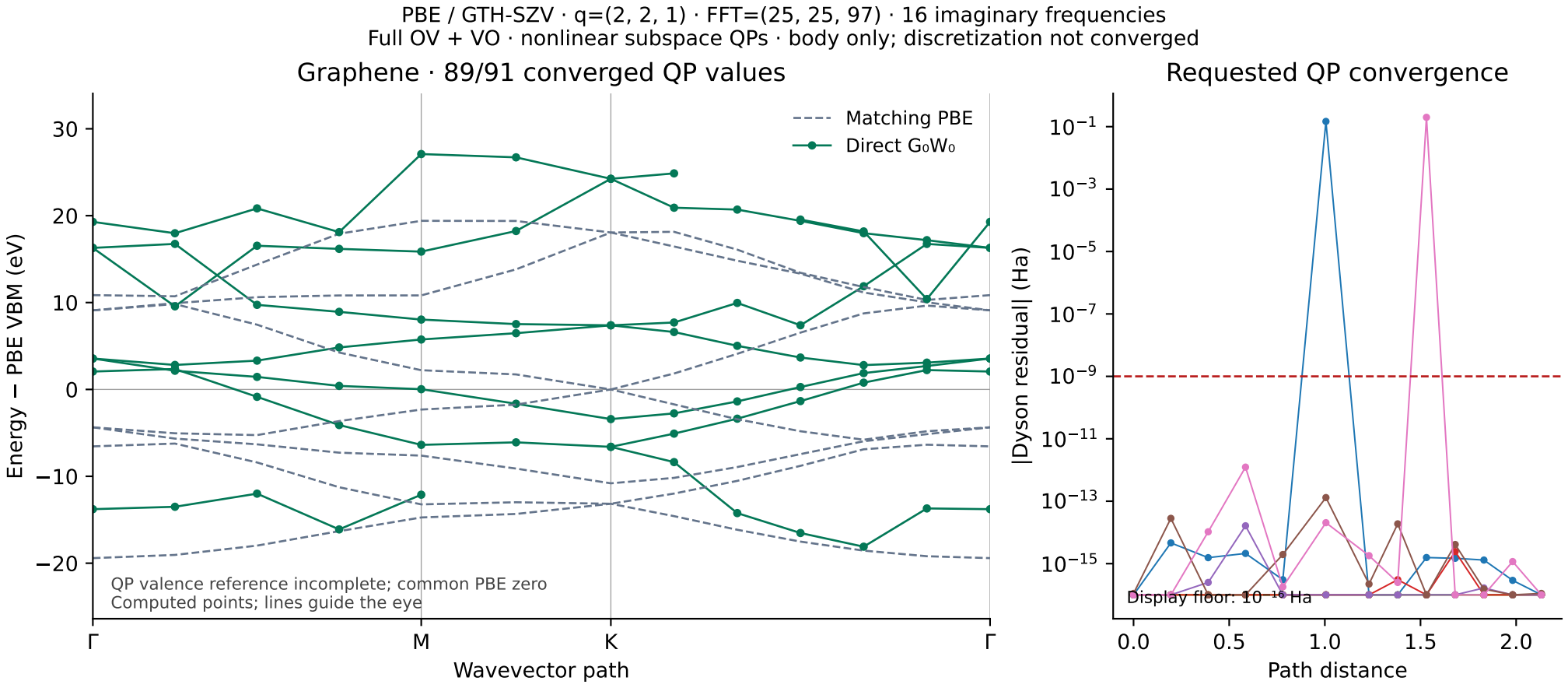
PBE / GTH-SZV · q=(2, 2, 1) · FFT=(25, 25, 97) · 16 imaginary frequencies
Full OV + VO · nonlinear subspace QPs · body only; discretization not converged
Graphene · 89/91 converged QP values
Matching PBE
Direct G₀W₀
Energy − PBE VBM (eV)
30
20
10
0
−10
−20
QP valence reference incomplete; common PBE zero
Computed points; lines guide the eye
Γ
M
K
Γ
Wavevector path
Requested QP convergence
|Dyson residual| (Ha)
10⁻¹
10⁻³
10⁻⁵
10⁻⁷
10⁻⁹
10⁻¹¹
10⁻¹³
10⁻¹⁵
Display floor: 10⁻¹⁶ Ha
0.0
0.5
1.0
1.5
2.0
Path distance